

DD-Mamba: A Forecast-Decomposable Dual-Domain State-Space Framework for Multivariate Time-Series Forecasting

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Abstract

Long-term multivariate time-series forecasting supports planning and decision making in areas such as energy, weather monitoring, and finance. Accurate prediction requires modeling temporal evolution, periodic structure, and dependence among variables, yet many hybrid architectures entangle these sources before a forecast is formed. We propose DD-Mamba, a forecast-decomposable dual-domain framework whose temporal and frequency branches each output a complete horizon forecast before a per-variable convex gate combines them. The temporal branch starts from a decomposition-based linear anchor and learns a zero-initialized correction, instantiated with a causal selective state-space encoder or a temporal multilayer perceptron according to the benchmark recipe. The frequency branch applies a complex-linear map that retains real–imaginary coupling while densely mixing Fourier bins and produces a standalone spectral forecast; selected configurations add a FITS-style spectral anchor. High-channel configurations share one bidirectional variate mixer across the branches to limit parameter growth. Across the twelve main benchmarks, DD-Mamba has the lowest reported average mean squared error (MSE) and mean absolute error (MAE) on nine datasets. Controlled component substitutions and removals characterize dataset-dependent effects, while matched hardware measurements quantify the associated training-time and GPU-memory costs.

Keywords: Time series forecasting, State-space models, Mamba, Frequency domain, Multivariate analysis

1. Introduction

Multivariate time series underpin planning and operation in electricity systems, weather monitoring, and financial markets [1, 2]. Given a finite observation window, a forecasting model must produce future trajectories for many interacting variables, and its error depends not only on the prediction horizon but also on temporal scale, distribution shift, and the dependence among channels. An effective model therefore needs to preserve local dynamics, capture repeated or slowly varying structure, and use information from other variables when it is relevant.

Forecasting methodology has evolved along this line. Classical statistical models such as ARIMA capture linear dependence under explicit assumptions [3]. Deep forecasters then relaxed these assumptions: DeepAR learns a global autoregressive probabilistic model across related series, and the Temporal Fusion Transformer combines recurrent local processing with attention and variable selection [4, 5]. Transformer-based models introduce decomposition or frequency-enhanced attention [6, 7], patch-based and inverted attention change the temporal unit or the role of variables [8, 9], and TimesNet reshapes the sequence into multi-period tensors to capture intra-period variation [10]. DLinear provides a lightweight decomposition-based alternative [11], while selective state-space models provide near-linear sequence modeling for long contexts [12, 13]. These developments show that temporal evolution, spectral

structure, and inter-variable dependence are all valuable, but they are often coupled inside a single latent representation.

Such coupling creates an architectural limitation. In many hybrid models, temporal and spectral features interact before either pathway forms a forecast. Although this can improve representation capacity, early feature fusion makes the information flow difficult to inspect and leaves the role of each domain implicit. The same issue arises when channel interaction is distributed throughout an encoder: cross-variable processing becomes inseparable from temporal modeling, even when dependencies among variables are weak or dense mixing is costly. This motivates a central question: how can a forecasting architecture combine temporal dynamics, periodic structure, and cross-variable relations while keeping the contribution of each pathway explicit?

We argue that domain complementarity does not require latent entanglement. A temporal pathway and a frequency pathway can each preserve forecast semantics, with their outputs combined only at the prediction stage. This organization makes the architecture forecast-decomposable: the two branch predictions and their fusion weights retain explicit meanings, and cross-variable interaction occupies a defined part of the information flow. The branches can therefore specialize in different structures while maintaining a traceable connection to the final prediction.

To this end, we propose DD-Mamba, whose overall architecture is illustrated in Fig. 1. The temporal pathway couples a linear trend–seasonal anchor with a zero-initialized nonlinear

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correction. A causal selective state-space encoder supplies this correction in the ETT, Weather, Solar-Energy, and Exchange recipes, whereas high-channel ECL and PEMS configurations use a temporal multilayer perceptron with variable-axis state-space mixing. The frequency pathway operates on the complex fast Fourier transform coefficients of real-valued inputs, preserves real–imaginary coupling, and densely mixes Fourier bins before producing a standalone forecast. A FITS-style spectral anchor is enabled only in selected configurations [14]. A bidirectional variate mixer can model cross-channel relations before the forecasting heads; high-channel recipes tie one mixer across both branches. Finally, a learned convex gate blends the two complete branch forecasts at the prediction level.

Contributions. (1) We introduce a dual-domain architecture in which the temporal and spectral pathways each produce a complete horizon forecast and a per-variable convex gate combines them only at the prediction level. (2) We develop two domain-specific forecast paths: a decomposition-based linear temporal anchor with a zero-initialized learnable correction, and a complex-linear spectral encoder that retains real–imaginary coupling while densely mixing Fourier bins before predicting the horizon. The correction encoder and optional FITS-style spectral anchor are selected by benchmark recipe rather than assumed to be universal. (3) For high-channel datasets, we tie one bidirectional variate-Mamba mixer across both branches while retaining branch-specific prediction heads, providing a parameter-efficient form of cross-variable modeling. Together, these choices yield a forecast-decomposable network whose component effects can be tested directly, although the present experiments support dataset-dependent rather than uniform gains.

The remainder of the paper is organized as follows. Section 2 reviews related work on classical, Transformer-based, state-space, and spectral forecasting. Section 3 presents DD-Mamba. Section 4 reports the experimental results and efficiency analysis. Section 5 discusses the findings and limitations, and Section 6 concludes the paper.

2. Related Work

Research on long-term time-series forecasting has developed along several complementary directions, including deep sequence architectures, lightweight predictors, frequency-aware methods, spatial–temporal dependency modeling, and state-space models. These directions differ mainly in how they represent temporal evolution, periodic structure, and dependence among variables [15].

Deep forecasting architectures. Recurrent forecasters learn temporal dynamics sequentially, as represented by the global autoregressive formulation of DeepAR and the recurrent-attention design of the Temporal Fusion Transformer [4, 5]. Transformer-based studies extend the receptive field through sparse attention, decomposition and auto-correlation, or frequency-enhanced blocks [16, 6, 7]. Other architectures reorganize the forecasting tokens: PatchTST models local temporal patches, iTransformer treats variable histories as tokens, and TimesNet maps a one-dimensional series into multi-period

representations [8, 9, 10]. These approaches increase the capacity to represent long contexts, but temporal resolution, periodic processing, and inter-variable interaction are often embedded in the same backbone. The resulting information flow is therefore difficult to trace, and attention-based interaction can remain expensive for long sequences or high-dimensional inputs.

Linear and lightweight forecasting. DLinear shows that decomposition-based linear projections can provide a strong long-horizon baseline with low architectural complexity [11]. TiDE introduces residual nonlinear mappings, TimeMixer performs decomposable multiscale mixing, and SparseTSF uses cross-period sparse modeling to obtain a lightweight predictor [17, 18, 19]. Their shared advantage is computational simplicity. However, these designs usually represent different temporal structures within one projection space and do not explicitly coordinate time- and frequency-domain forecasts.

Module-level and recipe-oriented evaluation. Recent studies have shifted attention from monolithic model rankings to the contribution and stability of individual forecasting components. TimeRecipe systematically benchmarks module-level choices across datasets and task settings and uses the resulting evidence to recommend scenario-specific recipes, while CombinationTS decomposes forecasting pipelines into orthogonal modules and evaluates their expected performance and stability under shared conditions [20, 21]. These frameworks motivate controlled attribution rather than assuming that one backbone or recipe is uniformly preferable. DD-Mamba follows this evaluation perspective at the architecture level: its experiments replace or remove temporal, spectral, normalization, fusion, and variable-mixing components, while its methodological focus remains the forecast-level organization of two complete domain-specific predictions rather than an exhaustive search over the full module space.

Frequency-aware and dual-domain forecasting. Frequency-aware methods share the premise that periodic patterns can be represented compactly in a spectral or multiresolution space. Direct spectral methods learn transformations over Fourier coefficients [22, 14, 23], while dual-domain networks combine temporal and spectral information through convolutional analysis, hierarchical representation learning, or wavelet decomposition [24, 25, 26]. Related decomposition methods separate trend, seasonal, dispersion, or multiscale components before forecasting [6, 27, 28]. Despite their different operators, these methods often introduce spectral information inside an intermediate encoder or fuse it with temporal features before either pathway forms a prediction. This early coupling promotes feature interaction, but it also makes the domain-specific route to the final forecast implicit. A fixed decomposition or spectral selection also prespecifies the form of periodic structure exposed to later layers, which may limit adaptation when that structure changes over time.

Spatial–temporal and state-space forecasting. Spatial–temporal models jointly encode temporal evolution and inter-variable structure. Attention-based methods construct dependencies from variable tokens or dimension segments, whereas fusion, convolutional, and graph-learning designs exchange information through a global core, temporal-channel kernels,

multiview graphs, or adaptive global–local graphs [29, 9, 30, 31, 32, 33]. These methods differ in whether spatial dependence is dense, sparse, or dynamic, but they share the aim of propagating information among variables. Indiscriminate interaction may also transmit irrelevant signals when useful dependencies are weak or time varying. Selective state-space models offer a linear-complexity alternative for long-sequence processing: Mamba uses input-dependent recurrent updates to control how information is retained [12]. For time-series forecasting, S-Mamba applies bidirectional state-space scans over variable tokens, TimeMachine combines channel-independent and channel-mixing representations at multiple scales, and Dec-Mamba integrates state-space modeling with trend–seasonal decomposition [13, 34, 35]. Existing state-space forecasters therefore provide efficient temporal or spatial interaction, while explicit spectral modeling is usually developed along a separate architectural path.

Taken together, prior studies provide effective mechanisms for temporal encoding, periodic representation, and spatial interaction, but these mechanisms are commonly concentrated in a single backbone or coupled at an intermediate feature level. A remaining architectural challenge is to coordinate temporal refinement, spectral regularity, and inter-variable dependence while keeping their forecasting paths explicit. DD-Mamba addresses this issue with a decomposition-anchored temporal path, a complex-linear frequency path that retains real–imaginary coupling, optional variable-axis state-space mixing, and prediction-level gated fusion of two complete forecasts. The novelty lies in this forecast-level organization and its testable component semantics, not in claiming the first use of Fourier transforms, complex-valued layers, or state-space models for forecasting.

3. Methodology

3.1. Preliminaries

Problem formulation. Let $\mathbf{X} = [\mathbf{x}_1, \dots, \mathbf{x}_L]^\top \in \mathbb{R}^{L \times C}$ denote a multivariate history, where L is the look-back length, C is the number of variables, and $\mathbf{x}_t \in \mathbb{R}^C$ is the observation at time step t . Given $\mathbf{X}$, the forecasting task is to estimate the next H observations, $\mathbf{Y} = [\mathbf{x}_{L+1}, \dots, \mathbf{x}_{L+H}]^\top \in \mathbb{R}^{H \times C}$. The model therefore learns a mapping

$$\hat{\mathbf{Y}} = \mathcal{F}_\theta(\mathbf{X}), \quad \mathcal{F}_\theta : \mathbb{R}^{L \times C} \rightarrow \mathbb{R}^{H \times C}, \quad (1)$$

where θ contains all trainable parameters and $\hat{\mathbf{Y}}$ is the predicted future sequence.

Selective state-space model. For one feature channel, a continuous-time state-space model maps an input $u(t)$ to an output $y(t)$ through a latent state $\mathbf{h}(t) \in \mathbb{R}^N$:

$$\dot{\mathbf{h}}(t) = \mathbf{A}\mathbf{h}(t) + \mathbf{B}u(t), \quad y(t) = \mathbf{C}^\top \mathbf{h}(t) + Du(t), \quad (2)$$

where $\mathbf{A} \in \mathbb{R}^{N \times N}$ is the state-transition matrix, $\mathbf{B}, \mathbf{C} \in \mathbb{R}^N$ are the input and output projections, and D is a skip coefficient. Under a zero-order hold with step size $\Delta_k > 0$, the exact discrete

update is

$$\begin{aligned} \bar{\mathbf{A}}_k &= \exp(\Delta_k \mathbf{A}), \\ \bar{\mathbf{B}}_k &= \int_0^{\Delta_k} \exp((\Delta_k - \tau)\mathbf{A}) \mathbf{B}_k d\tau, \\ \mathbf{h}_k &= \bar{\mathbf{A}}_k \mathbf{h}_{k-1} + \bar{\mathbf{B}}_k u_k, \quad y_k = \mathbf{C}_k^\top \mathbf{h}_k + Du_k. \end{aligned} \quad (3)$$

Mamba makes Δ_k , $\mathbf{B}_k$, and $\mathbf{C}_k$ functions of the current token, while retaining a structured state matrix $\mathbf{A}$ [12]. The resulting selective scan adapts state retention and input injection to the sequence content and evaluates the recurrence with complexity linear in the sequence length. When selected as the temporal correction encoder in DD-Mamba, this operator is scanned causally along time. The variable mixer, when enabled, instead scans the unordered variable axis in both directions.

3.2. Overall architecture

The overall framework of DD-Mamba is shown in Fig. 1. It contains two parallel forecast paths with different input representations. The temporal branch operates on the sampled sequence and couples a decomposition-based linear forecast with a learned residual correction. The correction encoder is causal Mamba in the ETT, Weather, Solar-Energy, and Exchange recipes and a temporal MLP in the high-channel ECL and PEMS recipes. The frequency branch transforms the same history into the Fourier domain, densely mixes its complex coefficients, and forms a separate horizon forecast. Spatial interaction is confined to a bidirectional variable mixer placed before the branch heads. The two paths remain separate until their forecasts are combined by a learned convex gate, preserving an explicit temporal forecast, an explicit frequency forecast, and a variable-wise fusion weight.

When the reversible instance normalization (RevIN) wrapper is enabled, it maps $\mathbf{X}$ to a normalized sequence $\tilde{\mathbf{X}}$ independently for each variable and applies the corresponding inverse transformation to the fused prediction [36]. This wrapper surrounds the forecasting core and does not alter the definitions of its branches. All projections, state-space parameters, mixers, and forecast heads are optimized jointly from scratch; no external encoder or pretrained checkpoint is loaded. Initialization coordinates the two paths. The temporal correction head is initialized to zero, so its initial branch output equals the linear anchor. When a FITS-style frequency anchor is present, the spectral anchor and its correction head also start at zero. Otherwise the frequency head retains its standard random initialization. The final gate starts with zero weights and bias 2.2, yielding $\sigma(2.2) \simeq 0.90$ and a time-dominant initial mixture.

3.3. Temporal branch

The temporal branch first separates the normalized history into a slowly varying trend and a residual seasonal component. For an odd moving-average width $w = 2q + 1$, define

$$\begin{aligned} \mathcal{M}_w(\tilde{\mathbf{X}})_{t,c} &= \frac{1}{w} \sum_{j=-q}^q \tilde{X}_{\pi(t+j),c}, \\ \mathbf{X}_{\text{tr}} &= \mathcal{M}_w(\tilde{\mathbf{X}}), \quad \mathbf{X}_{\text{se}} = \tilde{\mathbf{X}} - \mathbf{X}_{\text{tr}}, \end{aligned} \quad (4)$$

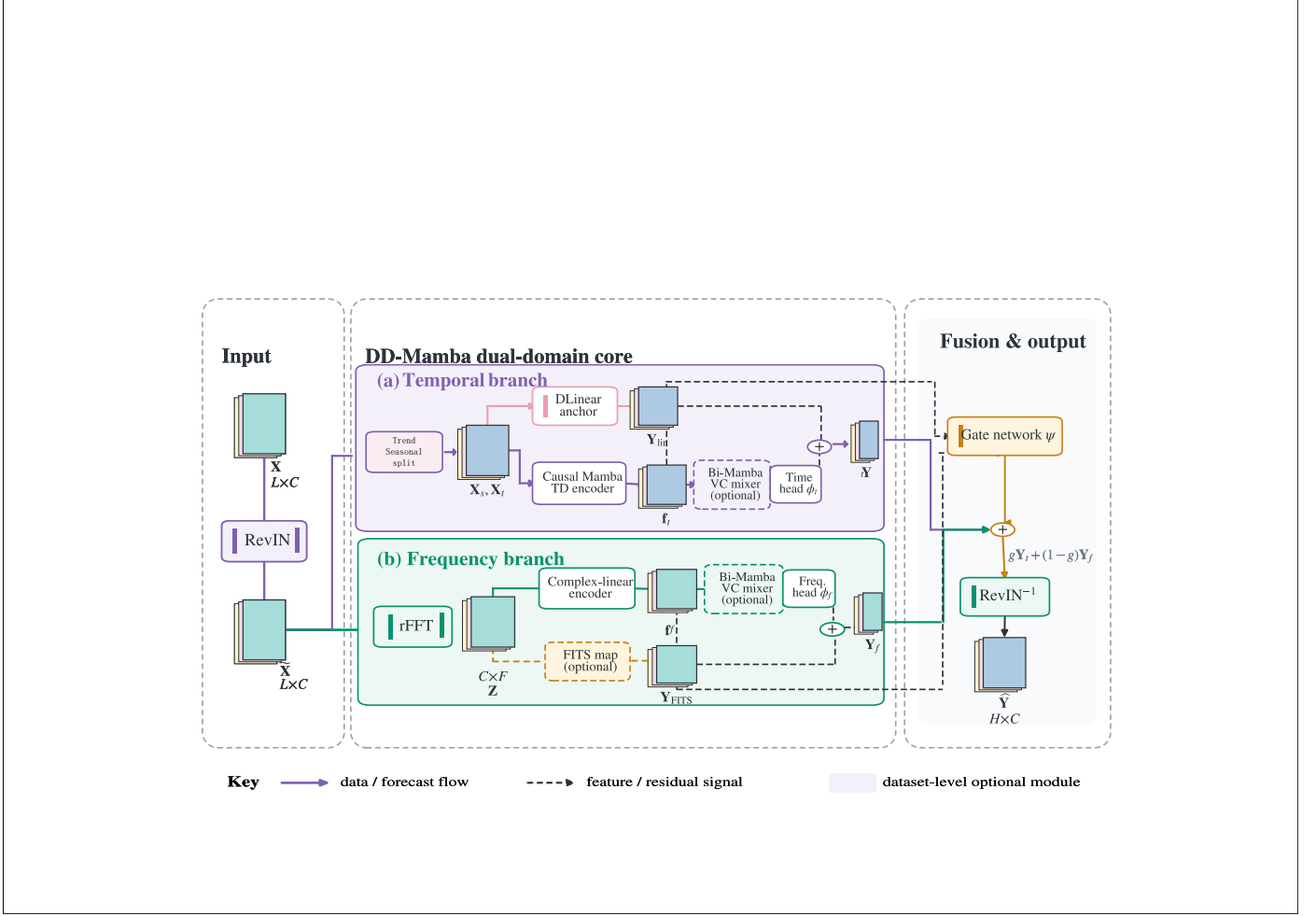


Figure 1: Overall architecture of DD-Mamba. The temporal branch combines a trend–seasonal linear anchor with a zero-initialized correction; the diagram shows its causal Mamba instantiation, while high-channel recipes use a temporal MLP. The frequency branch applies a dense complex-linear encoder and, on selected datasets, an additional FITS-style spectral anchor. Bidirectional Mamba mixers model interactions across variables, and a convex gate combines the two complete branch forecasts before inverse normalization.

where $\pi(\cdot)$ clips an index to $[1, L]$, which is equivalent to replication padding at the sequence boundaries. Thus, $\mathbf{X}_{\text{tr}}$ represents the local level retained by the moving average, while $\mathbf{X}_{\text{se}}$ contains the shorter-scale deviations around that level.

A DLinear-style anchor maps the two components separately from the look-back window to the forecast horizon [11]:

$$\mathbf{Y}_{\text{lin}}[:, c] = \mathbf{W}_{\text{tr}}\mathbf{X}_{\text{tr}}[:, c] + \mathbf{W}_{\text{se}}\mathbf{X}_{\text{se}}[:, c], \quad c = 1, \dots, C, \quad (5)$$

where $\mathbf{W}_{\text{tr}}, \mathbf{W}_{\text{se}} \in \mathbb{R}^{H \times L}$ are shared across variables. The separate projections allow the slowly varying level and the residual fluctuations to contribute directly to the horizon-level prediction.

For each variable, the paired trend and seasonal histories are embedded as a length- L latent sequence and processed by a correction encoder $\mathcal{E}_{\theta}^{\text{time}}$. The standard low-channel recipes instantiate $\mathcal{E}_{\theta}^{\text{time}}$ as a causal Mamba encoder and summarize its last state. High-channel recipes instantiate it as a residual temporal MLP followed by mean pooling. A forecasting head maps the

resulting feature to a residual correction:

$$\begin{aligned} \mathbf{f}_{\text{time}}^{(c)} &= \mathcal{E}_{\theta}^{\text{time}}(\mathbf{X}_{\text{tr}}[:, c], \mathbf{X}_{\text{se}}[:, c]), \\ \mathbf{Y}_{\text{time}}[:, c] &= \mathbf{Y}_{\text{lin}}[:, c] + \phi_{\text{time}}(\mathbf{f}_{\text{time}}^{(c)}). \end{aligned} \quad (6)$$

The linear anchor provides the direct trend–seasonal forecast, while the correction path models deviations not captured by the two linear projections. The last linear layer in ϕ_{time} is initialized to zero. Consequently, $\mathbf{Y}_{\text{time}} = \mathbf{Y}_{\text{lin}}$ at initialization, and the nonlinear correction is introduced progressively during training. This anchor–refinement construction is shared by all reported configurations; the choice between causal Mamba and a temporal MLP is dataset dependent.

3.4. Frequency branch

Let $K = \lfloor L/2 \rfloor + 1$. The frequency branch applies the real-valued fast Fourier transform along time, $\mathbf{Z} = \text{rFFT}_L(\tilde{\mathbf{X}}^T) \in \mathbb{C}^{C \times K}$. A retention operator $\mathcal{R}_{\rho}$ may remove the highest fraction ρ of bins, giving $\mathbf{Z}_{\rho} = \mathcal{R}_{\rho}(\mathbf{Z}) \in \mathbb{C}^{C \times K_{\rho}}$. Write $\mathbf{Z}_{\rho} = \mathbf{Z}_r + i\mathbf{Z}_i$ and

$\mathbf{W} = \mathbf{W}_r + i\mathbf{W}_i$. The complex-linear encoder computes

$$\mathbf{O}_r = \mathbf{Z}_r \mathbf{W}_r^\top - \mathbf{Z}_i \mathbf{W}_i^\top, \quad \mathbf{O}_i = \mathbf{Z}_r \mathbf{W}_i^\top + \mathbf{Z}_i \mathbf{W}_r^\top. \quad (7)$$

The dense matrices mix all retained Fourier bins while preserving real-imaginary coupling. Concatenating $\mathbf{O}_r$ and $\mathbf{O}_i$ and applying a real-valued projection yields the branch feature $\mathbf{F}_{\text{freq}} \in \mathbb{R}^{C \times d}$. The encoder is used in every reported recipe and is distinct from the optional FITS-style anchor below.

The optional FITS-style path maps the retained spectrum to the frequency grid of the concatenated history and forecast. Let $K' = \lfloor (L+H)/2 \rfloor + 1$ and $\mathbf{U} \in \mathbb{C}^{K' \times K_\rho}$. Its time-domain forecast is

$$\mathbf{Y}_{\text{FITS}}^\top = \Pi_H \left[\text{irFFT}_{L+H}(\mathbf{Z}_\rho \mathbf{U}^\top) \right], \quad (8)$$

where $\Pi_H : \mathbb{R}^{C \times (L+H)} \rightarrow \mathbb{R}^{C \times H}$ retains the final H reconstructed samples. The complete frequency forecast is

$$\mathbf{Y}_{\text{freq}}^\top = \mathbf{Y}_{\text{FITS}}^\top + \phi_{\text{freq}}(\mathbf{F}_{\text{freq}}), \quad (9)$$

where ϕ_{freq} maps the spectral feature to the horizon. If the interpolation path is omitted, $\mathbf{Y}_{\text{FITS}}$ is set to zero and the complex-linear encoder with its forecast head forms the complete branch output. The operator in Eq. (8) follows the frequency-interpolation formulation of FITS [14]; it is enabled only for ETTh1, ETTh2, and Weather. Thus, the dense complex encoder is the common spectral representation, whereas the FITS-style map is a recipe-specific forecasting anchor.

3.5. Spatial-temporal modeling and forecast fusion

Temporal dependence is modeled by the correction encoder in Eq. (6). Spatial dependence refers to interactions among the C variables. For either branch $b \in \{\text{time}, \text{freq}\}$, stack the variable features as $\mathbf{F}_b \in \mathbb{R}^{C \times d}$. The bidirectional variable mixer is defined as

$$\text{BiM}_b(\mathbf{F}_b) = \mathcal{M}_b^+(\mathbf{F}_b) + \text{Rev}[\mathcal{M}_b^-(\text{Rev}(\mathbf{F}_b))], \quad (10)$$

where $\mathcal{M}_b^+$ and $\mathcal{M}_b^-$ are Mamba scans over variable tokens and $\text{Rev}(\cdot)$ reverses their order [13]. The two directions are combined because the variable axis has no causal orientation. When the mixer is present, its output is supplied to the corresponding branch head; otherwise, the branch feature passes through unchanged. Weather and Solar-Energy use separate mixers in the two branches. ECL and PEMS03/04/07/08 instead tie the mixer parameters, so BiM_{time} and BiM_{freq} refer to the same module. This shared placement reduces mixer parameters without tying the domain-specific forecast heads. The ETT and Exchange recipes omit the mixer.

Let $\tilde{\mathbf{F}}_{\text{time}}$ and $\tilde{\mathbf{F}}_{\text{freq}}$ denote the features delivered to the two forecast heads. A row-wise gate network produces one weight per variable,

$$\begin{aligned} \mathbf{g} &= \sigma(\psi([\tilde{\mathbf{F}}_{\text{time}}; \tilde{\mathbf{F}}_{\text{freq}}])) \in (0, 1)^C, \\ \mathbf{G} &= \mathbf{1}_H \mathbf{g}^\top, \\ \hat{\mathbf{Y}}_n &= \mathbf{G} \odot \mathbf{Y}_{\text{time}} + (\mathbf{1} - \mathbf{G}) \odot \mathbf{Y}_{\text{freq}}, \end{aligned} \quad (11)$$

where $[\cdot]$ denotes feature concatenation, σ is the sigmoid function, and $\odot$ denotes element-wise multiplication. Since each

Table 1: Dataset characteristics.

Datasets	Variates	Timesteps	Granularity
ETTh1/ETTh2	7	17,420	1 hour
ETTm1/ETTm2	7	69,680	15 min
Weather	21	52,696	10 min
Solar-Energy	137	52,560	10 min
ECL	321	26,304	1 hour
Exchange	8	7,588	1 day
PEMS03	358	26,208	5 min
PEMS04	307	16,992	5 min
PEMS07	883	28,224	5 min
PEMS08	170	17,856	5 min

entry of $\mathbf{G}$ lies between zero and one, Eq. (11) is a convex combination of two complete forecasts. This prediction-level fusion retains the semantic identity of both branches and exposes their learned weights directly. When normalization is enabled, the fused output is mapped back to the original scale by the inverse transformation defined above; otherwise, $\hat{\mathbf{Y}} = \hat{\mathbf{Y}}_n$.

4. Experiments

4.1. Experimental details

Experimental environment. All experiments are implemented in PyTorch 2.11 and conducted on a single NVIDIA GeForce RTX 5090 GPU. Models are optimized with AdamW using the mean squared error objective. Across tasks, the batch size ranges from 16 to 32, the initial learning rate from 10^{-4} to 10^{-3} , the weight decay from 10^{-4} to 5×10^{-4} , and the training budget from 10 to 30 epochs. Early stopping is based on validation loss, and architectural settings are selected on the validation set. The normalization wrapper is disabled on Solar-Energy according to validation performance.

Datasets and metrics. We evaluate DD-Mamba on ETTh1, ETTh2, ETTm1, ETTm2, Weather, Solar-Energy, ECL, Exchange, PEMS03, PEMS04, PEMS07, and PEMS08. The look-back length is $L = 96$; forecast horizons are $H \in \{96, 192, 336, 720\}$ for the long-term benchmarks and $H \in \{12, 24, 48, 96\}$ for the PEMS benchmarks. Table 1 reports the dataset statistics. Forecasting accuracy is measured by mean squared error (MSE) and mean absolute error (MAE), and dataset-level averages are computed over the displayed horizons.

Baselines. The comparison covers eleven representative methods: S-Mamba and ms-Mamba for state-space modeling [13, 37]; iTransformer, PatchTST, Crossformer, FEDformer, and Autoformer for Transformer-based forecasting [9, 8, 29, 7, 6]; DLinear, TiDE, and TimeMixer for linear or MLP-based forecasting [11, 17, 18]; and TimesNet for convolutional modeling [10]. ms-Mamba is used only in the efficiency comparison and TimeMixer only in the case study. Values reproduced from prior publications are identified in the corresponding table captions; comparisons based on mixed result sources are treated as descriptive.

Evaluation protocol. Per-horizon DD-Mamba results in the main tables are obtained locally, whereas the published baseline entries are reproduced from the sources identified in the table captions. Because the main tables combine local results with values reported in prior publications, their rankings are interpreted descriptively. The component and efficiency comparisons follow their respective common protocols. Independent provenance-complete replication remains necessary.

For the descriptive channel-dependence analysis, the mean absolute pairwise Pearson correlation is computed as

$$|\bar{r}| = \frac{2}{C(C-1)} \sum_{i < j} |\text{corr}(\mathbf{x}^{(i)}, \mathbf{x}^{(j)})|, \quad (12)$$

where $\mathbf{x}^{(i)}$ denotes the observed sequence of the i th variable.

4.2. Main Results

Across the twelve main benchmarks, DD-Mamba has the lowest reported average MSE and MAE on ETTh1, ETTh2, ETTm1, ETTm2, Weather, ECL, Exchange, Solar-Energy, and PEMS08. S-Mamba leads both averages on PEMS04 and PEMS07, while iTransformer leads both on PEMS03. These rankings are descriptive because the tables combine local DD-Mamba results with values reported in prior publications.

The broad performance across ETT, ECL, and Solar is consistent with the anchor-and-correction design: the decomposed linear path preserves trend and seasonal structure, the selected temporal encoder learns residual corrections, and the separate spectral path represents periodic structure. The convex gate combines these forecasts without forcing one domain to dominate every dataset. Because the high-channel recipes use a temporal MLP rather than causal Mamba, their results support the common anchor-refinement organization but do not constitute evidence for a universal state-space temporal encoder. The mixed PEMS rankings likewise show that the variate mixer is not uniformly advantageous. The controlled component analysis below examines these mechanisms directly on ETTh2, Weather, and Solar-Energy.

4.3. Component analysis

Following the substitution-and-removal protocol of S-Mamba [13], each variant changes one component while retaining the remaining configuration. The ETTh2, Weather, and Solar-Energy recipes use causal Mamba as the temporal-domain (TD) correction encoder, so the MLP is a controlled replacement in Table 5. This comparison should not be confused with the reported high-channel recipes, where the MLP is the selected full-model temporal encoder. The other replacement variants modify the frequency-domain (FD) encoder, fusion rule, variate-correlation (VC) encoder, or normalization setting; the removal variants omit the FD branch, TD branch, DLinear anchor, or VC mixer. The VC comparisons replace bi-Mamba with attention or remove the mixer. These comparisons are reported for Weather and Solar-Energy because the ETTh2 configuration contains no VC-mixer layer.

Table 5 shows that reversing the selected normalization setting produces the largest and most consistent deterioration, especially on ETTh2 at longer horizons and on Weather. Weather also benefits from the temporal and cross-variable pathways: replacing temporal Mamba with an MLP, removing the TD branch, replacing the VC encoder with attention, or removing the VC mixer increases its horizon-average error.

In contrast, replacing the FD encoder or learned fusion and removing the FD branch or DLinear anchor yield smaller or mixed changes, and several ETTh2 and Solar-Energy variants are close to or slightly better than the full model. The controlled contrasts therefore show dataset-dependent contributions from normalization, temporal correction, and cross-variable mixing, but they do not establish a universal component ranking beyond the evaluated configurations.

Table 2: ETT forecasting results. Published S-Mamba values are reproduced from Wang et al. [13], Table 3; the remaining published baseline values are reproduced from Liu et al. [9], Appendix F.4, Table 10.

Dataset	H	DD-Mamba	S-Mamba	iTransformer	PatchTST	Crossformer	TiDE	DLinear	FEDformer	Autoformer
		MSE MAE	MSE MAE	MSE MAE	MSE MAE	MSE MAE	MSE MAE	MSE MAE	MSE MAE	MSE MAE
ETTh1	96	<u>0.381</u> 0.397	0.386 0.405	0.386 0.405	0.414 0.419	0.423 0.448	0.479 0.464	0.386 <u>0.400</u>	0.376 0.419	0.449 0.459
	192	<u>0.429</u> 0.423	0.443 0.437	0.441 0.436	0.460 0.445	0.471 0.474	0.525 0.492	0.437 <u>0.432</u>	0.420 0.448	0.500 0.482
	336	<u>0.472</u> 0.446	0.489 0.468	0.487 <u>0.458</u>	0.501 0.466	0.570 0.546	0.565 0.515	0.481 0.459	0.459 0.465	0.521 0.496
	720	0.472 0.468	0.502 0.489	0.503 0.491	<u>0.500</u> <u>0.488</u>	0.653 0.621	0.594 0.558	0.519 0.516	0.506 0.507	0.514 0.512
	Avg	0.439 0.434	0.455 0.450	0.454 <u>0.447</u>	0.469 0.454	0.529 0.522	0.541 0.507	0.456 0.452	<u>0.440</u> 0.460	0.496 0.487
ETTh2	96	0.292 0.345	<u>0.296</u> <u>0.348</u>	0.297 0.349	0.302 <u>0.348</u>	0.745 0.584	0.400 0.440	0.333 0.387	0.358 0.397	0.346 0.388
	192	0.368 0.392	<u>0.376</u> <u>0.396</u>	0.380 0.400	0.388 0.400	0.877 0.656	0.528 0.509	0.477 0.476	0.429 0.439	0.456 0.452
	336	0.415 0.427	<u>0.424</u> <u>0.431</u>	0.428 0.432	0.426 0.433	1.043 0.731	0.643 0.571	0.594 0.541	0.496 0.487	0.482 0.486
	720	0.423 0.442	<u>0.426</u> <u>0.444</u>	0.427 0.445	0.431 0.446	1.104 0.763	0.874 0.679	0.831 0.657	0.463 0.474	0.515 0.511
	Avg	0.374 0.402	<u>0.381</u> <u>0.405</u>	0.383 0.407	0.387 0.407	0.942 0.684	0.611 0.550	0.559 0.515	0.437 0.449	0.450 0.459
ETTm1	96	0.321 0.359	0.333 0.368	0.334 0.368	<u>0.329</u> <u>0.367</u>	0.404 0.426	0.364 0.387	0.345 0.372	0.379 0.419	0.505 0.475
	192	0.366 0.385	0.376 0.390	0.377 0.391	<u>0.367</u> 0.385	0.450 0.451	0.398 0.404	<u>0.380</u> <u>0.389</u>	0.426 0.441	0.553 0.496
	336	0.396 0.405	0.408 0.413	0.426 0.420	<u>0.399</u> <u>0.410</u>	0.532 0.515	0.428 0.425	0.413 0.413	0.445 0.459	0.621 0.537
	720	<u>0.461</u> <u>0.443</u>	0.475 0.448	0.491 0.459	0.454 0.439	0.666 0.589	0.487 0.461	0.474 0.453	0.543 0.490	0.671 0.561
	Avg	0.386 0.398	0.398 0.405	0.407 0.410	<u>0.387</u> <u>0.400</u>	0.513 0.496	0.419 0.419	0.403 0.407	0.448 0.452	0.588 0.517
ETTm2	96	<u>0.177</u> <u>0.260</u>	0.179 0.263	0.180 0.264	0.175 0.259	0.287 0.366	0.207 0.305	0.193 0.292	0.203 0.287	0.255 0.339
	192	0.241 0.302	<u>0.250</u> <u>0.309</u>	<u>0.250</u> <u>0.309</u>	0.241 0.302	0.414 0.492	0.290 0.364	0.284 0.362	0.269 0.328	0.281 0.340
	336	0.302 0.341	<u>0.312</u> <u>0.349</u>	<u>0.311</u> <u>0.348</u>	<u>0.305</u> <u>0.343</u>	0.597 0.542	0.377 0.422	0.369 0.427	0.325 0.366	0.339 0.372
	720	0.398 0.397	0.411 0.406	0.412 0.407	<u>0.402</u> <u>0.400</u>	1.730 1.042	0.558 0.524	0.554 0.522	0.421 0.415	0.433 0.432
	Avg	0.280 0.325	0.288 0.332	0.288 0.332	<u>0.281</u> <u>0.326</u>	0.757 0.610	0.358 0.404	0.350 0.401	0.305 0.349	0.327 0.371

Notes: Bold red and underlined purple denote the best and second-best displayed values, respectively; ties receive the same mark.

Table 3: General-benchmark forecasting results. Published S-Mamba values are reproduced from Wang et al. [13], Table 4; the remaining published baseline values are reproduced from Liu et al. [9], Appendix F.4, Table 10.

Dataset	H	DD-Mamba	S-Mamba	iTransformer	PatchTST	Crossformer	TiDE	DLinear	FEDformer	Autoformer
		MSE MAE	MSE MAE	MSE MAE	MSE MAE	MSE MAE	MSE MAE	MSE MAE	MSE MAE	MSE MAE
Weather	96	<u>0.162</u> 0.207	0.165 <u>0.210</u>	0.174 0.214	0.177 0.218	0.158 0.230	0.202 0.261	0.196 0.255	0.217 0.296	0.266 0.336
	192	<u>0.212</u> <u>0.254</u>	0.214 0.252	0.221 <u>0.254</u>	0.225 0.259	0.206 0.277	0.242 0.298	0.237 0.296	0.276 0.336	0.307 0.367
	336	0.267 0.293	0.274 0.297	0.278 <u>0.296</u>	0.278 0.297	<u>0.272</u> 0.335	0.287 0.335	0.283 0.335	0.339 0.380	0.359 0.395
	720	<u>0.349</u> <u>0.347</u>	0.350 0.345	0.358 <u>0.347</u>	0.354 0.348	0.398 0.418	0.351 0.386	0.345 0.381	0.403 0.428	0.419 0.428
	Avg	0.247 0.275	<u>0.251</u> <u>0.276</u>	0.258 0.278	0.259 0.281	0.259 0.315	0.271 0.320	0.265 0.317	0.309 0.360	0.338 0.382
ECL	96	<u>0.140</u> 0.235	0.139 0.235	0.148 <u>0.240</u>	0.181 0.270	0.219 0.314	0.237 0.329	0.197 0.282	0.193 0.308	0.201 0.317
	192	0.157 0.252	<u>0.159</u> 0.255	0.162 <u>0.253</u>	0.188 0.274	0.231 0.322	0.236 0.330	0.196 0.285	0.201 0.315	0.222 0.334
	336	0.174 <u>0.271</u>	<u>0.176</u> 0.272	0.178 0.269	0.204 0.293	0.246 0.337	0.249 0.344	0.209 0.301	0.214 0.329	0.231 0.338
	720	0.200 0.296	<u>0.204</u> <u>0.298</u>	0.225 0.317	0.246 0.324	0.280 0.363	0.284 0.373	0.245 0.333	0.246 0.355	0.254 0.361
	Avg	0.168 0.263	<u>0.170</u> <u>0.265</u>	0.178 0.270	0.205 0.290	0.244 0.334	0.251 0.344	0.212 0.300	0.214 0.327	0.227 0.338
Exchange	96	0.082 0.200	<u>0.086</u> 0.207	<u>0.086</u> 0.206	0.088 <u>0.205</u>	0.256 0.367	0.094 0.218	0.088 0.218	0.148 0.278	0.197 0.323
	192	0.172 0.295	0.182 0.304	<u>0.177</u> <u>0.299</u>	<u>0.176</u> <u>0.299</u>	0.470 0.509	0.184 0.307	<u>0.176</u> 0.315	0.271 0.315	0.300 0.369
	336	<u>0.312</u> <u>0.403</u>	0.332 0.418	0.331 0.417	0.301 0.397	1.268 0.883	0.349 0.431	0.313 0.427	0.460 0.427	0.509 0.524
	720	0.762 0.654	0.867 0.703	0.847 <u>0.691</u>	0.901 0.714	1.767 1.068	0.852 0.698	<u>0.839</u> 0.695	1.195 0.695	1.447 0.941
	Avg	0.332 0.388	0.367 0.408	0.360 <u>0.403</u>	0.367 0.404	0.940 0.707	0.370 0.413	<u>0.354</u> 0.414	0.519 0.429	0.613 0.539
Solar-Energy	96	0.181 <u>0.241</u>	0.205 0.244	<u>0.203</u> 0.237	0.234 0.286	0.310 0.331	0.312 0.399	0.290 0.378	0.242 0.342	0.884 0.711
	192	0.189 0.256	0.237 0.270	<u>0.233</u> <u>0.261</u>	0.267 0.310	0.734 0.725	0.339 0.416	0.320 0.398	0.285 0.380	0.834 0.692
	336	0.207 0.269	0.258 0.288	<u>0.248</u> <u>0.273</u>	0.290 0.315	0.750 0.735	0.368 0.430	0.353 0.415	0.282 0.376	0.941 0.723
	720	0.206 0.269	0.260 0.288	<u>0.249</u> <u>0.275</u>	0.289 0.317	0.769 0.765	0.370 0.425	0.356 0.413	0.357 0.427	0.882 0.717
	Avg	0.196 0.259	0.240 0.273	<u>0.233</u> <u>0.262</u>	0.270 0.307	0.641 0.639	0.347 0.417	0.330 0.401	0.291 0.381	0.885 0.711

Notes: Bold red and underlined purple denote the best and second-best displayed values, respectively; ties receive the same mark. Solar-Energy uses the validation-selected normalization-off configuration (Section 4.1).

Table 4: PEMS forecasting results. Published baseline values are reproduced from Wang et al. [13], Table 2, with comparator entries following Liu et al. [9], Appendix F.4, Tables 9–10.

Dataset	H	DD-Mamba	S-Mamba	iTransformer	PatchTST	Crossformer	TiDE	DLinear	FEDformer	Autoformer
		MSE MAE	MSE MAE	MSE MAE	MSE MAE	MSE MAE	MSE MAE	MSE MAE	MSE MAE	MSE MAE
PEMS03	12	0.065 0.168	<u>0.065 0.169</u>	<u>0.071</u> 0.174	0.099 0.216	0.090 0.203	0.178 0.305	0.122 0.243	0.126 0.251	0.272 0.385
	24	0.085 0.193	<u>0.087 0.196</u>	0.093 0.201	0.142 0.259	0.121 0.240	0.257 0.371	0.201 0.317	0.149 0.275	0.334 0.440
	48	<u>0.128 0.238</u>	0.133 0.243	0.125 0.236	0.211 0.319	0.202 0.317	0.379 0.463	0.333 0.425	0.227 0.348	1.032 0.782
	96	<u>0.187 0.292</u>	0.201 0.305	0.164 0.275	0.269 0.370	0.262 0.367	0.490 0.539	0.457 0.515	0.348 0.434	1.031 0.796
	Avg	<u>0.116 0.223</u>	0.122 0.228	0.113 0.221	0.180 0.291	0.169 0.281	0.326 0.419	0.278 0.375	0.213 0.327	0.667 0.601
PEMS04	12	0.075 0.179	<u>0.076 0.180</u>	0.078 0.183	0.105 0.224	0.098 0.218	0.219 0.340	0.148 0.272	0.138 0.262	0.424 0.491
	24	<u>0.093 0.200</u>	0.084 0.193	0.095 0.205	0.153 0.275	0.131 0.256	0.292 0.398	0.224 0.340	0.177 0.293	0.459 0.509
	48	<u>0.123 0.233</u>	0.115 0.224	<u>0.120 0.233</u>	0.229 0.339	0.205 0.326	0.409 0.478	0.355 0.437	0.270 0.368	0.646 0.610
	96	<u>0.171 0.276</u>	0.137 0.248	<u>0.150 0.262</u>	0.291 0.389	0.402 0.457	0.492 0.532	0.452 0.504	0.341 0.427	0.912 0.748
	Avg	0.116 0.222	0.103 0.211	<u>0.111 0.221</u>	0.195 0.307	0.209 0.314	0.353 0.437	0.295 0.388	0.231 0.337	0.610 0.590
PEMS07	12	0.059 0.155	<u>0.063 0.159</u>	0.067 0.165	0.095 0.207	0.094 0.200	0.173 0.304	0.115 0.242	0.109 0.225	0.199 0.336
	24	0.079 0.175	<u>0.081 0.183</u>	0.088 0.190	0.150 0.262	0.139 0.247	0.271 0.383	0.210 0.329	0.125 0.244	0.323 0.420
	48	<u>0.105 0.209</u>	0.093 0.192	0.110 0.215	0.253 0.340	0.311 0.369	0.446 0.495	0.398 0.458	0.165 0.288	0.390 0.470
	96	<u>0.139 0.244</u>	0.117 0.217	<u>0.139</u> 0.245	0.346 0.404	0.396 0.442	0.628 0.577	0.594 0.553	0.262 0.376	0.554 0.578
	Avg	<u>0.095 0.196</u>	0.089 0.188	0.101 0.204	0.211 0.303	0.235 0.315	0.380 0.440	0.329 0.395	0.165 0.283	0.367 0.451
PEMS08	12	0.072 0.174	<u>0.076 0.178</u>	0.079 0.182	0.168 0.232	0.165 0.214	0.227 0.343	0.154 0.276	0.173 0.273	0.436 0.485
	24	0.093 0.195	<u>0.104 0.209</u>	0.115 0.219	0.224 0.281	0.215 0.260	0.318 0.409	0.248 0.353	0.210 0.301	0.467 0.502
	48	0.137 0.232	<u>0.167 0.228</u>	0.186 0.235	0.321 0.354	0.315 0.355	0.497 0.510	0.440 0.470	0.320 0.394	0.966 0.733
	96	0.208 0.277	0.245 0.280	<u>0.221</u> 0.267	0.408 0.417	0.377 0.397	0.721 0.592	0.674 0.565	0.442 0.465	1.385 0.915
	Avg	0.127 0.220	<u>0.148 0.224</u>	0.150 0.226	0.280 0.321	0.268 0.307	0.441 0.464	0.379 0.416	0.286 0.358	0.814 0.659

Notes: Bold red and underlined purple denote the best and second-best displayed values for each metric, respectively; ties receive the same mark.

Table 5: Component substitution and removal results on ETTh2, Weather, and Solar-Energy.

Component	Config	H	ETTh2		Weather		Solar		Component	Config	H	ETTh2		Weather		Solar	
			MSE	MAE	MSE	MAE	MSE	MAE				MSE	MAE	MSE	MAE	MSE	MAE
<i>Full model</i>									<i>Replace</i>								
All components	Default	96	0.293	0.344	0.162	0.207	0.178	0.237	Normalization	Reverse	96	0.319	0.366	0.182	0.244	0.194	0.238
		192	0.370	0.392	0.212	0.253	0.198	0.259			192	0.442	0.442	0.261	0.311	0.229	0.267
		336	0.416	0.427	0.268	0.295	0.208	0.269			336	0.534	0.497	0.336	0.368	0.245	0.282
		720	0.425	0.442	0.352	0.350	0.216	0.277			720	0.868	0.649	0.438	0.428	0.247	0.284
<i>Replace</i>									<i>w/o</i>								
TD encoder	MLP	96	0.293	0.343	0.164	0.210	0.188	0.245	FD branch	w/o	96	0.292	0.344	0.163	0.209	0.187	0.248
		192	0.369	0.391	0.212	0.254	0.202	0.264			192	0.375	0.397	0.212	0.254	0.195	0.259
		336	0.419	0.428	0.271	0.297	0.208	0.269			336	0.419	0.430	0.270	0.297	0.206	0.270
		720	0.429	0.444	0.353	0.351	0.208	0.267			720	0.427	0.447	0.351	0.351	0.212	0.275
FD encoder	bi-Mamba	96	0.292	0.341	0.163	0.211	0.180	0.241	TD branch	w/o	96	0.290	0.341	0.163	0.209	0.188	0.248
		192	0.372	0.393	0.210	0.254	0.194	0.258			192	0.366	0.391	0.213	0.254	0.204	0.265
		336	0.418	0.429	0.272	0.299	0.207	0.273			336	0.413	0.426	0.272	0.296	0.211	0.271
		720	0.421	0.442	0.347	0.349	0.214	0.277			720	0.422	0.442	0.355	0.352	0.214	0.272
Fusion	Mean	96	0.288	0.340	0.162	0.208	0.186	0.242	DLinear anchor	w/o	96	0.291	0.343	0.164	0.208	0.185	0.243
		192	0.367	0.391	0.212	0.254	0.200	0.259			192	0.369	0.392	0.212	0.254	0.196	0.259
		336	0.413	0.427	0.270	0.296	0.205	0.267			336	0.418	0.429	0.270	0.296	0.211	0.276
		720	0.422	0.442	0.351	0.350	0.213	0.272			720	0.427	0.444	0.352	0.350	0.221	0.282
VC encoder	Attention	96	–	–	0.171	0.213	0.185	0.241	VC mixer	w/o	96	–	–	0.172	0.215	0.192	0.255
		192	–	–	0.218	0.256	0.202	0.265			192	–	–	0.223	0.258	0.209	0.272
		336	–	–	0.277	0.299	0.207	0.271			336	–	–	0.276	0.297	0.214	0.276
		720	–	–	0.355	0.350	0.206	0.268			720	–	–	0.353	0.349	0.216	0.275

Notes: All configurations use $L = 96$ and are evaluated at $H \in \{96, 192, 336, 720\}$. “Reverse” swaps the full model’s normalization choice. Solar-Energy uses the validation-selected normalization-off configuration (Section 4.1).

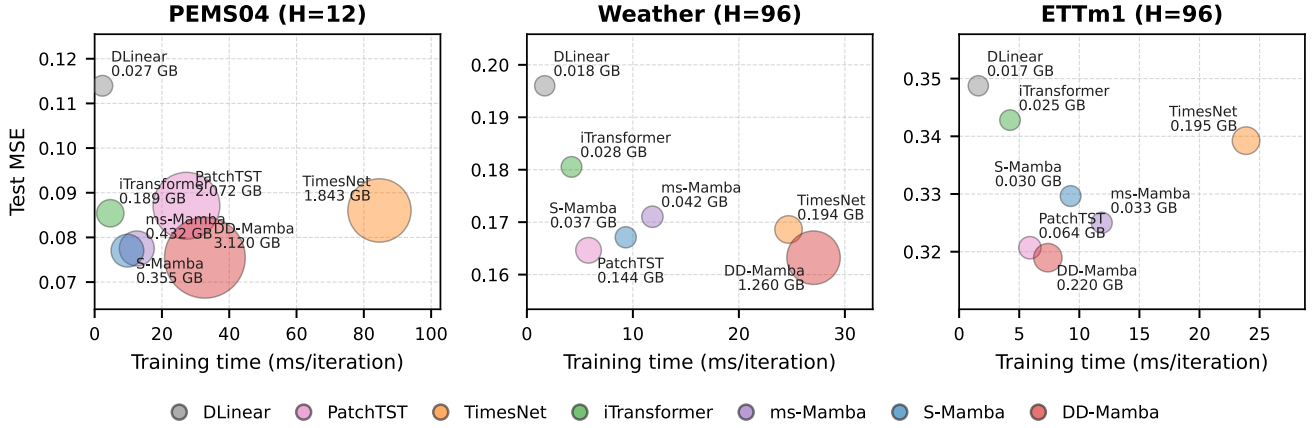


Figure 2: Matched training-efficiency comparison on PEMS04 ($H = 12$), Weather ($H = 96$), and ETTm1 ($H = 96$). Horizontal position is mean optimizer-loop time, vertical position is mean test MSE, and bubble area is linear in mean peak allocated GPU memory; direct annotations report peak memory in GB.

4.4. Matched training efficiency

The matched efficiency study compares test error, training time per optimizer iteration, and peak allocated GPU memory in the experimental environment defined above. Training time includes data loading, host-to-device transfer, forward and backward computation, and the optimizer update. Figure 2 visualizes the accuracy–time–memory trade-off, and Table 6 reports the corresponding means.

DD-Mamba has the lowest mean MSE in all three panels, but its resource profile depends strongly on dataset width. On PEMS04, it uses 32.77 ms per iteration and 3.120 GB, compared with 9.74 ms and 0.355 GB for the next-lowest-error S-Mamba. On Weather, its 27.04 ms and 1.260 GB likewise exceed PatchTST’s 5.82 ms and 0.144 GB. The compact ETTm1 configuration narrows the time gap (7.38 versus 5.89 ms for PatchTST) but still uses more peak memory (0.220 versus 0.064 GB); it remains faster than S-Mamba, ms-Mamba, and TimesNet. These results establish an accuracy–resource trade-off, not uniform computational superiority.

The largest memory gaps occur on PEMS04 and Weather, where the variate mixer is enabled (shared across branches on PEMS04 and separate by branch on Weather). ETTm1 omits the mixer but still executes two complete forecast paths, giving a smaller yet visible memory overhead relative to PatchTST. This pattern is consistent with cross-variable processing amplifying cost as channel count grows and dual-branch forecasting imposing a base overhead even at low channel counts. Weight sharing limits parameter growth but does not remove the activations or computation of repeated branch and variate operations. The experiment does not isolate the resource contribution of each module.

Table 6: Matched training-efficiency results corresponding to Fig. 2. All baseline values are measured in the same experimental environment.

Dataset	H	Model	Test MSE	Time (ms/iter.)	Peak memory (GB)
PEMS04	12	DLinear	0.11395	2.30	0.027
		PatchTST	0.08709	27.29	2.072
		TimesNet	0.08604	84.65	1.843
		iTransformer	0.08541	4.66	0.189
		ms-Mamba	0.07753	12.53	0.432
		S-Mamba	<u>0.07704</u>	9.74	0.355
		DD-Mamba	0.07546	32.77	3.120
Weather	96	DLinear	0.19601	1.68	0.018
		iTransformer	0.18053	4.21	0.028
		ms-Mamba	0.17103	11.84	0.042
		TimesNet	0.16861	24.69	0.194
		S-Mamba	0.16712	9.33	0.037
		PatchTST	<u>0.16463</u>	5.82	0.144
		DD-Mamba	0.16322	27.04	1.260
ETTM1	96	DLinear	0.34875	1.60	0.017
		iTransformer	0.34281	4.23	0.025
		TimesNet	0.33925	23.87	0.195
		S-Mamba	0.32965	9.28	0.030
		ms-Mamba	0.32503	11.87	0.033
		PatchTST	<u>0.32069</u>	5.89	0.064
		DD-Mamba	0.31897	7.38	0.220

Notes: Values are means obtained under the common efficiency protocol. Bold red and underlined purple identify the lowest and second-lowest test MSE in each panel; resource columns are not ranked.

4.5. Robustness analysis

To examine how DD-Mamba behaves when the deployed inputs are imperfect, we corrupt the standardized test windows of ETTh2 with zero-mean i.i.d. Gaussian noise of standard deviation σ and evaluate the trained models without any retraining or adaptation. Training and validation proceed exclusively on clean data, and the prediction targets are never corrupted, so the perturbation isolates the sensitivity of the learned forecast map to imperfect observations. We sweep $\sigma \in \{0.1, 0.2, 0.3, 0.5\}$ together with a clean anchor at $\sigma = 0$ across all four forecast

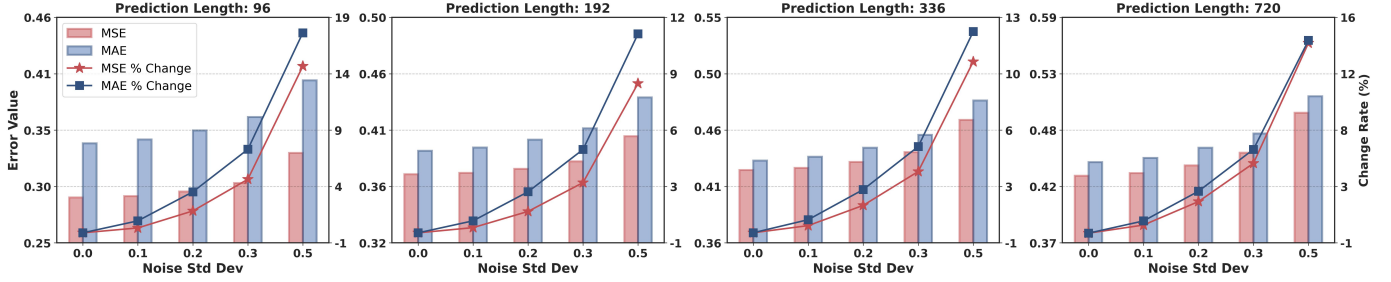


Figure 3: Robustness of DD-Mamba to input noise on ETTh2. Zero-mean Gaussian noise with standard deviation $\sigma \in \{0.1, 0.2, 0.3, 0.5\}$ is added to the standardized test inputs. Each panel reports one forecast horizon; bars show MSE and MAE on the left axis, and the markers trace the relative change with respect to the clean anchor $\sigma = 0$ on the right axis.

lengths, using the validation-selected configuration. The resulting trends are interpreted descriptively rather than as a matched baseline comparison.

Figure 3 shows that the degradation grows smoothly and monotonically with the noise level, without abrupt failure at any horizon. Under moderate corruption up to $\sigma = 0.3$, the MSE of DD-Mamba increases by at most 5.2 percent relative to the clean anchor, and the MAE remains within 7.4 percent. The penalty concentrates at the most severe setting: at $\sigma = 0.5$ the MSE rises from 0.290 to 0.333 at horizon 96 and from 0.434 to 0.495 at horizon 720, corresponding to relative degradations of 14.7 and 14.1 percent, while the largest MAE increase across horizons is 17.6 percent. Degradation does not grow uniformly with the forecast length; the horizon-192 cell is the most stable, with an MSE increase of only 8.3 percent even at $\sigma = 0.5$.

The gradual degradation is consistent with the availability of a decomposition-based temporal anchor. In parallel, the temporal correction and the complete spectral forecast remain active before the gate forms the final prediction. This organization offers a plausible explanation for the absence of abrupt failure as σ increases, but the self-only noise sweep neither separates the branch contributions nor compares their robustness with a matched baseline. The result is therefore descriptive.

4.6. Lookback analysis

Figure 4 reports validation-only sweeps for the high-dimensional ECL setting. Both panels use selected-checkpoint metrics from the same local lookback experiment. Among the five displayed models, DD-Mamba has the lowest validation MSE at the shortest history, Crossformer is lowest from $L = 48$ through $L = 480$, and PatchTST is lowest at the three longest histories. The curves therefore show architecture-dependent responses to additional context rather than a uniformly preferred lookback.

For DD-Mamba, extending the input from very short histories produces the largest reduction in both errors, while later gains are modest. The alternative models also respond differently: Crossformer is strongest across most short and medium histories, whereas PatchTST benefits most at the longest histories. Because the study covers one horizon and one seed, it is treated as descriptive evidence and is not used to revise the main test results.

4.7. Case study

Figure 5 exposes both shared forecasting structure and window-specific failure modes. On HUFL, the models recover the timing of the daytime decline and subsequent rise, but they compress the amplitude of the ground-truth trough. The discrepancy persists in the auxiliary day-3 panel at $H = 384$, where the black trajectory contains sharper local excursions than any prediction. On LUFL, the methods capture the broad fall-recovery pattern on day 2, while the auxiliary day-3 panel shows increasing phase and terminal-level errors. These panels therefore indicate a common smoothing bias around abrupt changes rather than a failure to recover the low-frequency shape.

The displayed trajectories do not uniformly favor DD-Mamba. It appears closest to the ground truth on the smoother LUFL day-2 segment. Among the other panels, TimeMixer appears closest on HUFL day 1 and LUFL day 3, whereas S-Mamba appears closest on HUFL day 3. These are qualitative readings of the rendered trajectories, not per-window error estimates or substitutes for the aggregate comparisons.

The accurate fall-recovery shape in the LUFL day-2 panel is consistent with the combination of a trend-seasonal linear anchor and a causal state-space correction, which preserves slow evolution while adapting the local trajectory. The shared smoothing around abrupt HUFL excursions suggests the complementary limitation: the linear anchor, spectral representation, and MSE objective all favor recurrent or low-frequency structure over isolated spikes. Thus the design offers a plausible explanation for both the successful shape tracking and the observed loss of amplitude; the four windows do not support a broader causal claim.

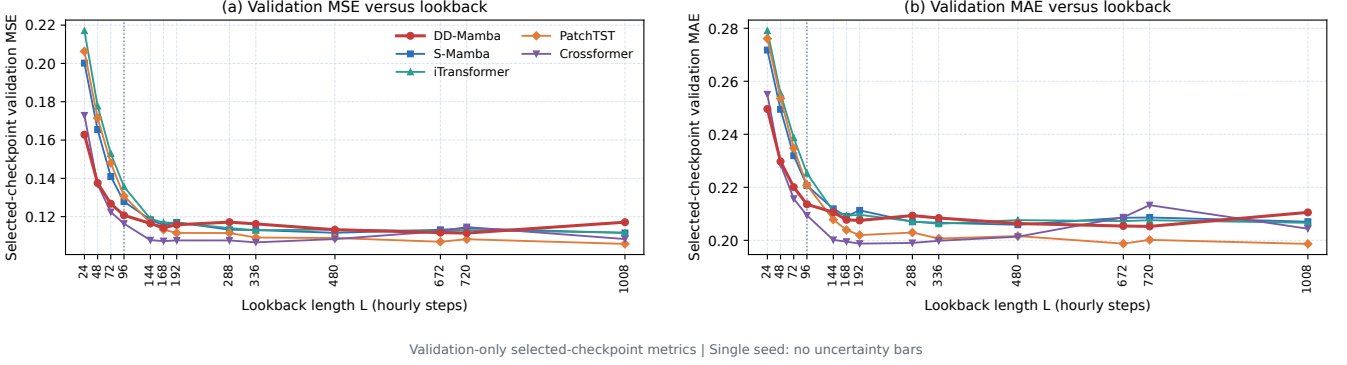


Figure 4: Validation-only lookback analysis on ECL at horizon 96. Panels (a)–(b) compare the selected-checkpoint MSE and MAE of DD-Mamba with S-Mamba, iTransformer, PatchTST, and Crossformer. All points are single-seed results, so no uncertainty bars are shown.

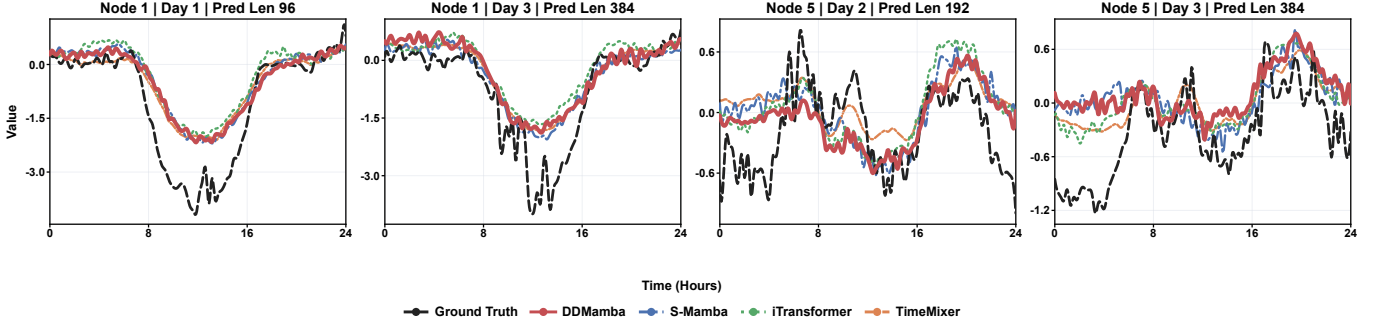


Figure 5: Qualitative ETTm1 forecasts for four selected 24-hour windows from sample 0. “Node 1” and “Node 5” correspond to the HUFL and LUFL channels, respectively. The black dashed line is the ground truth, the embedded legend label “DDMamba” denotes DD-Mamba, and the compared baselines are S-Mamba, iTransformer, and TimeMixer. Panel titles identify the displayed day and horizon; the auxiliary $H = 384$ panels are shown only to complete the third 24-hour segment and are excluded from all quantitative analyses. Curves are rendered with cubic-spline interpolation, which enters no quantitative comparison.

5. Discussion

5.1. Interpretation

The controlled component results are strongly dataset dependent. Reversing the selected normalization setting produces the largest and most consistent penalties, particularly on ETTh2 and Weather. Weather also benefits from its temporal and cross-variable pathways, whereas the frequency-encoder and fusion replacements yield smaller or mixed changes. Several ETTh2 and Solar-Energy variants remain close to or slightly better than the full model, so the results do not support a domain-independent component ranking.

The efficiency and lookback studies place the accuracy results in context. DD-Mamba has the lowest mean MSE in the PEMS04, Weather, and ETTm1 efficiency panels, but it is slower and uses more peak memory than the closest-error comparator on PEMS04 and Weather. Separately, the validation-only ECL comparison shows architecture-specific responses to longer histories: DD-Mamba is strongest at the shortest lookback, Crossformer across most short and medium histories, and PatchTST at the longest histories. These measurements do not establish uniform computational superiority or a universally preferred context length.

Because each component variant changes one mechanism while retaining the remaining recipe, Table 5 provides direct local contrasts for the evaluated configurations. Its three-dataset scope nevertheless limits broader attribution: the same replacement or removal need not behave identically on datasets outside ETTh2, Weather, and Solar-Energy.

The ETTm1 case study and dataset statistics further limit the interpretation. The displayed forecasts retain slow daily structure but smooth sharp excursions, and DD-Mamba is visually closest in only one of four selected windows. Channel correlation and time-varying scale help describe datasets but do not reliably determine whether a mixer or normalization should be used. Horizon-dependent errors, the neutral FITS-style result on ECL, and the ineffective longer training schedule are therefore retained as negative evidence rather than converted into broader design rules.

5.2. Threats to validity

Four limitations bound the findings. First, the baseline evidence mixes local reimplementations under a common protocol with values reported in prior publications, so tuning budgets, implementation fidelity, and uncertainty estimates are not uniform. The main tables therefore support descriptive rankings

rather than confirmatory performance claims. Second, the controlled component analysis covers three datasets and the specified replacement and removal choices. The ECL lookback sweep and selected ETTm1 windows are narrower diagnostic or qualitative analyses and should not be interpreted as population-level effects.

Third, the bidirectional variate mixer is not permutation-invariant, and no channel-order stress test is available. Fourth, the efficiency study covers only three dataset-horizon panels, one GPU type, and training-loop time; it does not measure inference latency, throughput, energy, or multi-GPU scaling. Stronger confirmation requires validation-only model selection, broader independent replication, complete run provenance, channel-order tests, and measurements across implementations and hardware environments.

6. Conclusion

In this paper, we propose DD-Mamba, a forecast-decomposable dual-domain framework whose temporal, spectral, normalization, and cross-variable components can be replaced or removed in controlled comparisons. A decomposition-based linear anchor with a zero-initialized learned correction forms the temporal path. The correction encoder is causal Mamba in the standard low-channel recipes and a temporal MLP in the high-channel recipes. A complex-linear encoder that retains real-imaginary coupling forms a complete frequency forecast, selected configurations add a FITS-style spectral anchor, and a convex gate merges the two branch predictions.

Across the twelve main benchmarks, DD-Mamba has the lowest reported average MSE and MAE on nine datasets: the four ETT datasets, Weather, ECL, Exchange, Solar-Energy, and PEMS08. The controlled component analysis shows the most consistent penalties when the selected normalization setting is reversed and supports the temporal and cross-variable pathways on Weather, while the fusion and frequency changes remain small or mixed. The matched efficiency study places these results in context: DD-Mamba attains the lowest MSE in the PEMS04, Weather, and ETTm1 panels, but incurs the largest training-time and memory costs on PEMS04 and Weather and a smaller overhead on ETTm1. The ECL validation lookback study shows that preferred context length depends on the architecture. These descriptive findings do not establish a universal ranking of the components.

Overall, DD-Mamba offers a compact, forecast-decomposable evaluation framework together with bounded, dataset-specific evidence about which mechanisms justify their computational cost. Future work includes lag-aware dependence modeling, broader independent replication, and validation-only component and hyperparameter selection.

CRedit authorship contribution statement

[Author 1 full name]: Conceptualization, Methodology, Software, Validation, Formal analysis, Investigation, Data curation, Visualization, Writing – original draft.

[Author 2 full name]: Methodology, Validation, Data curation, Writing – review & editing.

[Author 3 full name]: Supervision, Project administration, Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Declaration of generative AI and AI-assisted technologies in the manuscript preparation process

During the preparation of this work the author(s) used Anthropic Claude and OpenAI Codex to assist with typesetting, consistency checks, and language editing. After using these tools, the author(s) reviewed and edited the content as needed and take(s) full responsibility for the content of the published article.

Data availability

Data will be made available on request.

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